



Report on Task 2

**Title: Rice Variety Classification from Microscopic Images using CNN
with Attention**

Course Title: Machine Learning

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Group: 03

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1. Introduction

This report presents Task 2 of the Group03 project on the PRBD (Processed Rice Bangladesh Dataset). Building on the exploratory data analysis (Task 1), which confirmed a balanced 10-class dataset of 2000 original microscopic rice-grain images with no exploitable colour or brightness shortcut, this task covers two stages: (1) training and evaluating four representative CNN baselines under an identical, leakage-free protocol, and (2) designing, training, and evaluating a proposed attention-based model, ResNet50 combined with the Convolutional Block Attention Module (CBAM) to test whether attention improves rice-variety classification over the strongest baseline.

All models are trained and evaluated using the same class-stratified 70/10/20 train/validation/test split, built only on the Original_Images folder to avoid data leakage from near-duplicate augmented images. All reported metrics are computed on the held-out test set (400 images, 40 per class).

2. Data Split and Preprocessing

To prevent data leakage, the class-stratified 70/10/20 split was built once, using only the 2000 original images, and saved to a CSV file so that both the baseline notebook and the proposed model notebook load the identical split - no re-splitting, no risk of the same source image appearing in both training and test sets.

Split	Percentage	Images per class	Total images
Train	70%	140	1400
Validation	10%	20	200
Test	20%	40	400

Preprocessing applied identically to all models: resize to 224×224, ImageNet mean/std normalization, and (train split only) augmentation - horizontal/vertical flip, $\pm 20^\circ$ rotation, brightness/contrast jitter, hue/saturation jitter, and coarse dropout. Since Task 1's EDA confirmed the dataset is perfectly balanced (200 images per class), no class-imbalance handling (e.g. class weighting or oversampling) was required.

3. Baseline Models

Four representative baselines were trained and evaluated under the identical protocol described above, covering a from-scratch CNN and three widely used ImageNet-pretrained transfer-learning backbones:

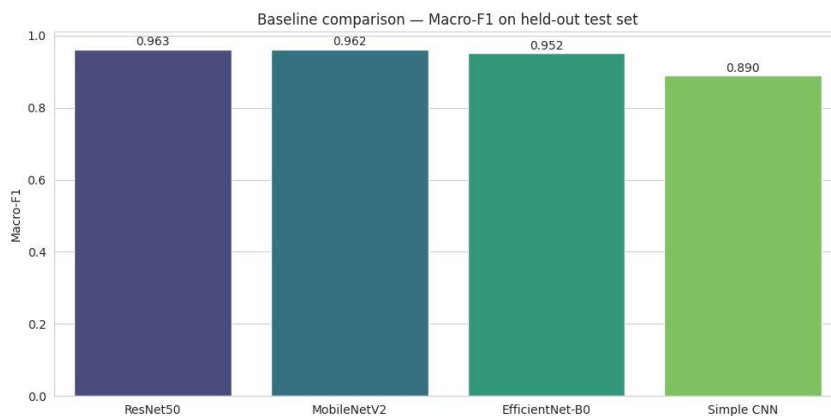
- Simple CNN - a small convolutional network trained entirely from scratch (no pretraining), serving as the from-scratch reference point.
- ResNet50 - ImageNet-pretrained (IMAGENET1K_V2 weights), fine-tuned on layer4 only.
- MobileNetV2 - a lightweight ImageNet-pretrained backbone, chosen to represent an efficient mobile-friendly baseline.
- EfficientNet-B0 - a modern, parameter-efficient ImageNet-pretrained backbone.

3.1 Baseline Evaluation Results (Test Set)

Model	Accuracy	Precision (M)	Recall (M)	F1 (M)	ROC-AUC (M)	Params	Size (MB)
Simple CNN	0.8900	0.8930	0.8900	0.8898	0.9936	423,562	1.63
ResNet50	0.9625	0.9654	0.9625	0.9625	0.9990	14,985,226	90.06
MobileNetV2	0.9625	0.9631	0.9625	0.9622	0.9983	1,538,890	8.77
EfficientNet-B0	0.9525	0.9540	0.9525	0.9524	0.9985	3,168,550	15.62

(M) = macro-average across all 10 classes.

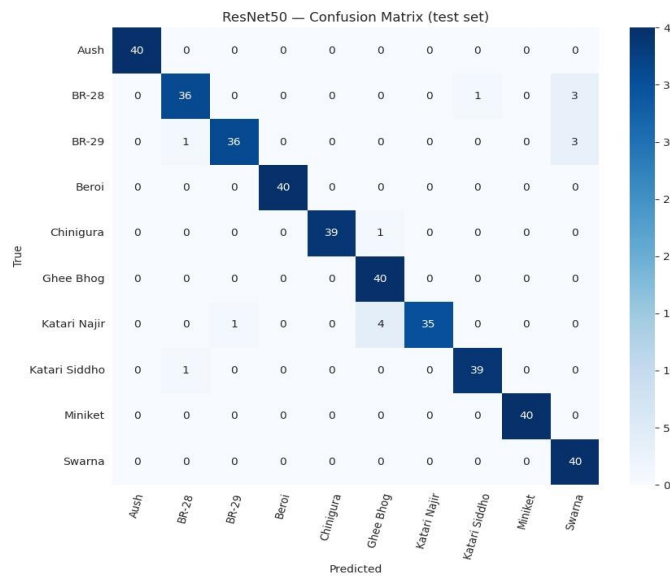
Baseline comparison bar chart - Macro-F1 across all 4 baselines:



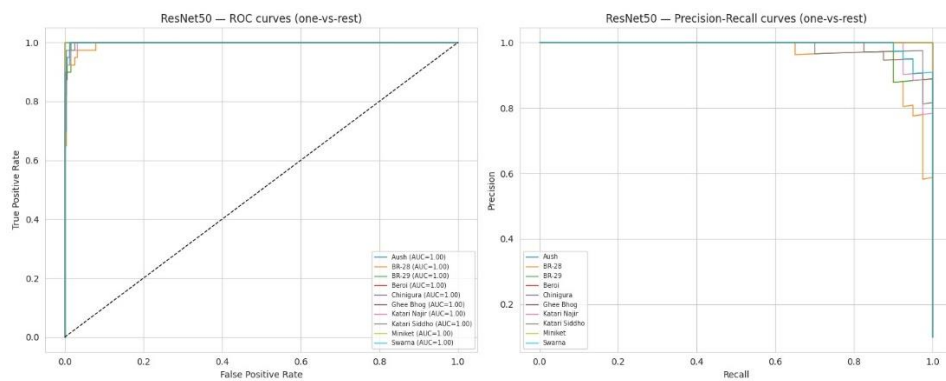
3.2 Which baseline is strongest?

ResNet50 is the strongest baseline on the test set (Macro-F1 = 0.9625), with MobileNetV2 essentially tied (0.9622). EfficientNet-B0 trails slightly (Macro-F1 = 0.9524), and the from-scratch Simple CNN is clearly weakest (Macro-F1 = 0.8898), showing that transfer learning alone already provides most of the achievable headroom on this dataset before any attention module is added. ResNet50's Macro-F1 = 0.9625 is therefore the exact bar that the proposed attention model must beat.

Confusion matrix - ResNet50 (best baseline, test set):



ROC and Precision-Recall curves - ResNet50 (best baseline, test set):



4. Proposed Model - ResNet50 + CBAM

The proposed model adds the Convolutional Block Attention Module (CBAM), a lightweight, hybrid channel-and-spatial attention mechanism to the strongest baseline backbone, ResNet50. ResNet50 was deliberately chosen as the backbone (rather than the weaker from-scratch Simple CNN) so that any change in performance can be attributed cleanly to attention, rather than being confounded with a weaker starting backbone.

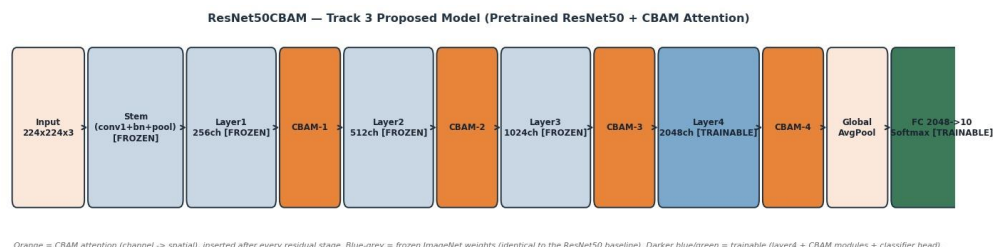
4.1 Why CBAM (Attention Type)

Task 1's EDA found that PRBD classes look visually very similar to each other (most varieties share the same elongated grain shape, and Beroi/Katari Najir/Katari Siddho are especially close). A plain ResNet50 head pools every channel and every spatial location equally, with no explicit mechanism to prioritize the small, subtle regions or channels that separate look-alike varieties. CBAM was chosen specifically to address this:

- Channel attention — answers "which feature channels are worth listening to?" using global average- and max-pooling, a shared MLP bottleneck, and a sigmoid gate.
- Spatial attention — answers "which pixel locations are worth looking at?" using channel-pooled average/max maps, concatenation, and a 7×7 convolution + sigmoid gate.
- CBAM chains the two (channel attention first, then spatial), the standard ordering from Woo et al., ECCV 2018.

This is a hybrid channel + spatial attention mechanism, inserted after every residual stage of ResNet50 (layer1 $\rightarrow$ 256ch, layer2 $\rightarrow$ 512ch, layer3 $\rightarrow$ 1024ch, layer4 $\rightarrow$ 2048ch), so the network can re-weight both channels and spatial locations at every scale, from fine low-level texture to coarse high-level shape, before the features reach the classifier head.

4.2 Architecture Diagram



4.3 Settings / Hyperparameters

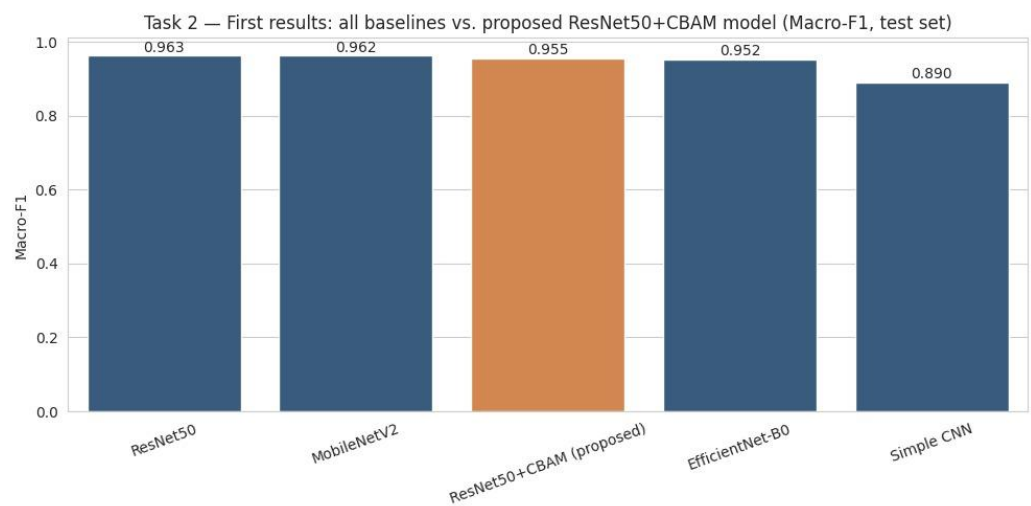
Setting	Value
Input size	$224 \times 224 \times 3$
Backbone	ResNet50, ImageNet IMAGENET1K_V2 pretrained weights
Fine-tuning budget	frozen: stem, layer1, layer2 — trainable: layer3, layer4 (layer3 unfrozen so the backbone can co-adapt with CBAM)
Attention	CBAM after layer1/2/3/4 (channel reduction ratio = 16, spatial kernel = 7×7) — always trainable
Classifier head	Dropout(0.3) $\rightarrow$ FC(2048 $\rightarrow$ 10)
Optimizer	Adam, weight_decay = $1e-4$
Learning rate	Differential: $5e-5$ for layer3/layer4 (matches the ResNet50 baseline's fine-tuning LR); $5e-4$ for the 4 CBAM modules + classifier head
LR schedule	ReduceLROnPlateau (factor 0.5, patience 2) on validation loss
Loss	Cross-entropy
Batch size	32
Max epochs	50, early stopping patience = 6 (validation loss)
Augmentation (train only)	H/V flip, $\pm 20^\circ$ rotation, brightness/contrast, hue/saturation, coarse dropout
Normalization	ImageNet mean/std (identical to all 4 baselines)
Split	Identical class-stratified 70/10/20 split loaded from the baselines notebook

Two intentional differences versus the ResNet50 baseline's recipe: (1) CBAM attention modules are added, and (2) layer3 is additionally unfrozen (beyond the baseline's layer4-only fine-tuning) so that the backbone's higher-level features can co-adapt jointly with the newly initialized CBAM modules a deliberate, documented deviation rather than a silent change.

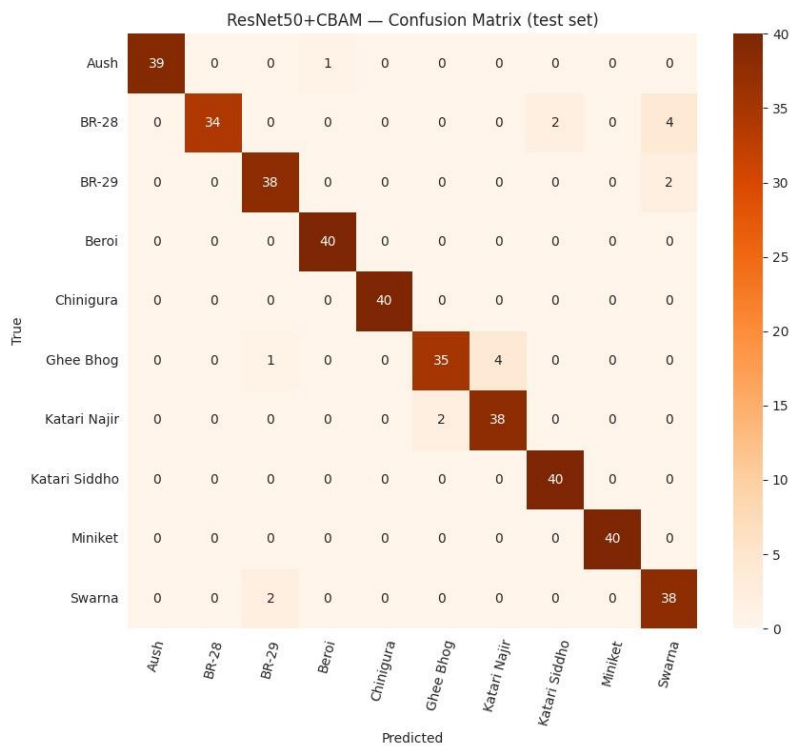
5. Does Attention Help?

Model	Accuracy	Precision (M)	Recall (M)	F1 (M)	ROC-AUC (M)	Params	Size (MB)
ResNet50 (baseline)	0.9625	0.9654	0.9625	0.9625	0.9990	14,985,226	90.06
MobileNetV2 (baseline)	0.9625	0.9631	0.9625	0.9622	0.9983	1,538,890	8.77
ResNet50 + CBAM (proposed)	0.9550	0.9569	0.9550	0.9548	0.9975	22,780,306	92.72
EfficientNet-B0 (baseline)	0.9525	0.9540	0.9525	0.9524	0.9985	3,168,550	15.62
Simple CNN (baseline)	0.8900	0.8930	0.8900	0.8898	0.9936	423,562	1.63

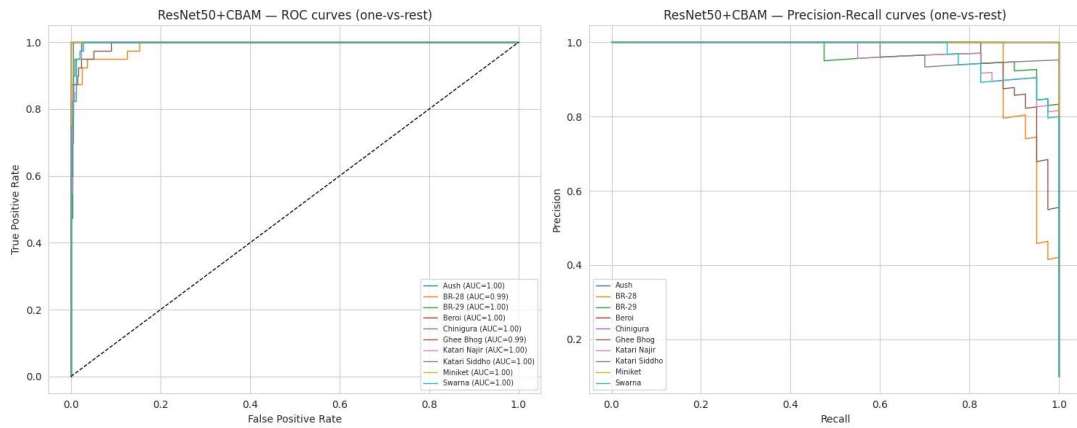
Final comparison bar chart - best baseline vs. proposed model:



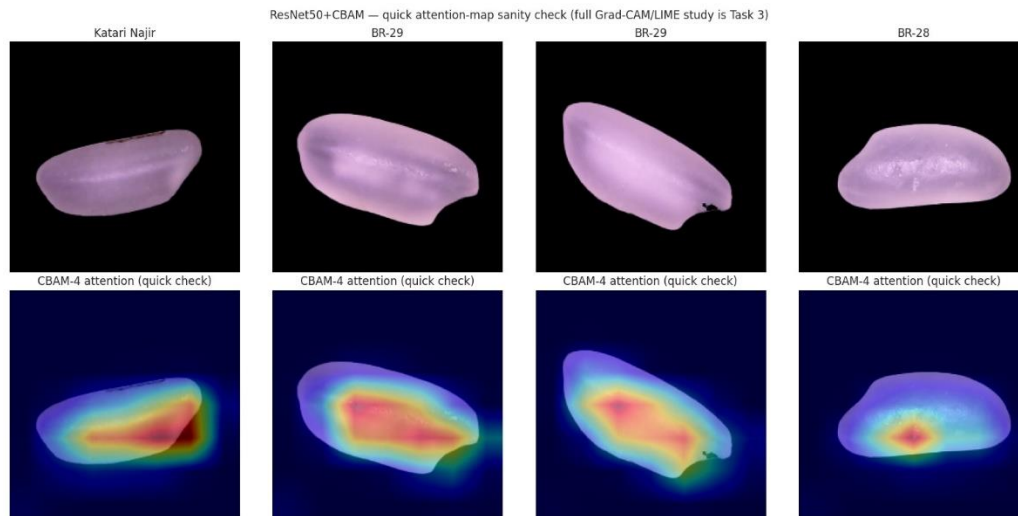
Confusion matrix - ResNet50+CBAM (proposed model):



ROC and Precision-Recall curves - ResNet50+CBAM (proposed model):



Qualitative CBAM spatial-attention map - one sample image with overlay:



5.1 Conclusion

On the same ResNet50 backbone and fine-tuning budget, adding CBAM attention gives Macro-F1 = 0.9548, 0.0077 lower than the plain ResNet50 baseline (0.9625). So in this first run, attention did not improve the strongest backbone; the proposed model ranks 3rd overall, behind ResNet50 and MobileNetV2, and just ahead of EfficientNet-B0.

This is the opposite of an earlier from-scratch experiment (not the main baseline reported above), where adding CBAM to a plain Simple CNN improved Macro-F1 by +0.0096. A plausible explanation is that ResNet50's ImageNet features are already strong and well-suited to this task, leaving CBAM comparatively little useful signal to re-weight, while the extra parameters (22.8M vs. 15.0M) and the joint fine-tuning of layer3 and layer4 together may make optimization harder within the same epoch and learning-rate budget.

This negative delta is reported as a legitimate, honest first result rather than being hidden or reframed. Task 2 does not require the proposed model to beat the baseline only to report first results honestly. Task 3's ablation study is exactly where this will be investigated further: for example, testing CBAM at only some residual stages, comparing channel-only vs. spatial-only attention, or reverting to the layer4-only fine-tuning budget used by the ResNet50 baseline to isolate whether the extra unfrozen layer, rather than CBAM itself, is responsible for the drop.